CHEM1043 Tutorial 1

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Topic 1 Gases: Properties and Behaviour

- Properties of gas: Gas pressure
- The simple gas laws
- The ideal gas equation and general gas equation
- Application of ideal gas equation
- Gas in chemical reactions
- Mixture of gases
- Kinetic-molecular theory of gas
- Non-ideal gas

Properties of gas: Gas Pressure

- Definition of pressure: force per unit area

$$P\,(Pa) = \frac{\text{Force (N)}}{\text{Area } (m^2)}$$

- Measuring indirectly by a liquid pressure

$$P\,(Pa) = \frac{F}{A} = \frac{W}{A} = \frac{m \times g}{A} = \frac{V \times d \times g}{A} = \frac{h \times A \times d \times g}{A} = h \times d \times g$$

weight is force

$W = m \times g$

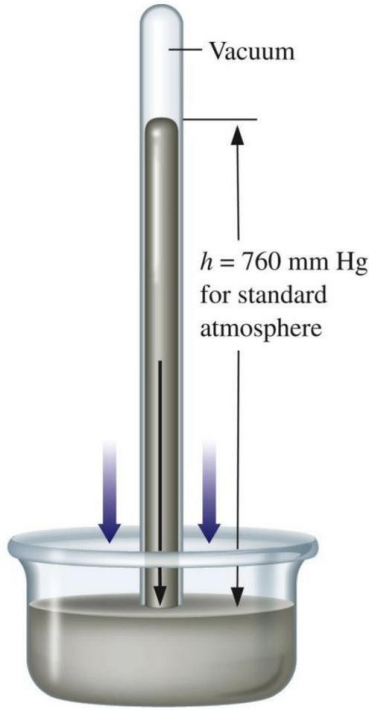
$m = V \times d$

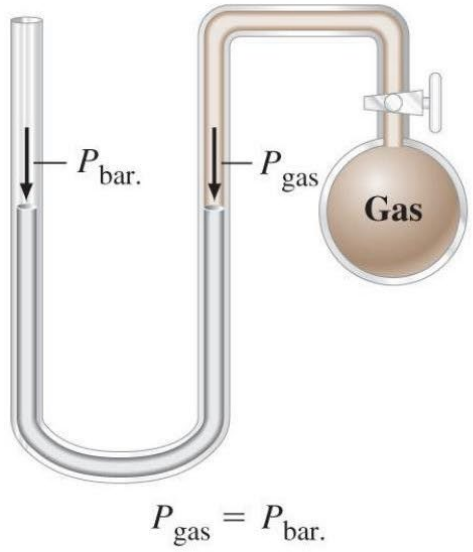
$V = A \times h$

g is constant

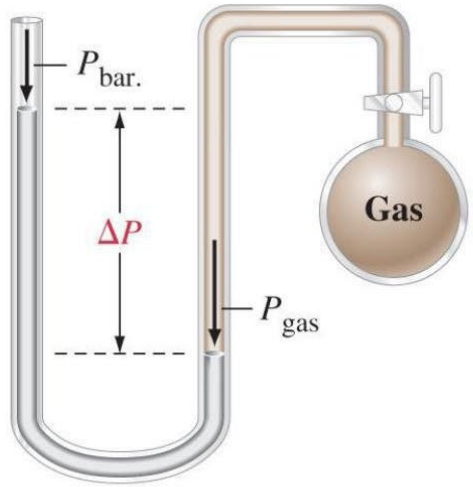
Properties of gas: Gas Pressure

- Barometer
- Manometer

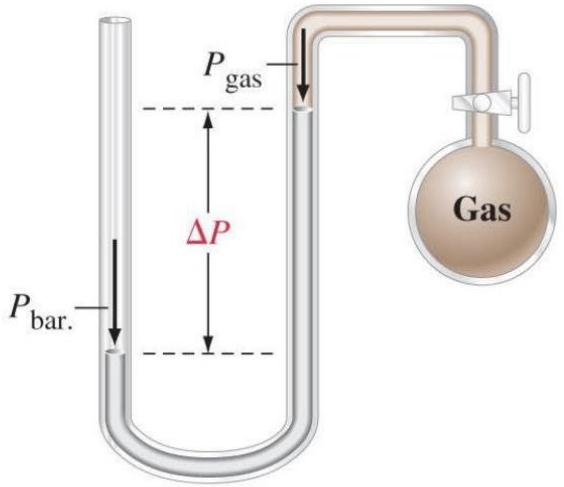




(a) Gas pressure equal to barometric pressure



(b) Gas pressure greater than barometric pressure

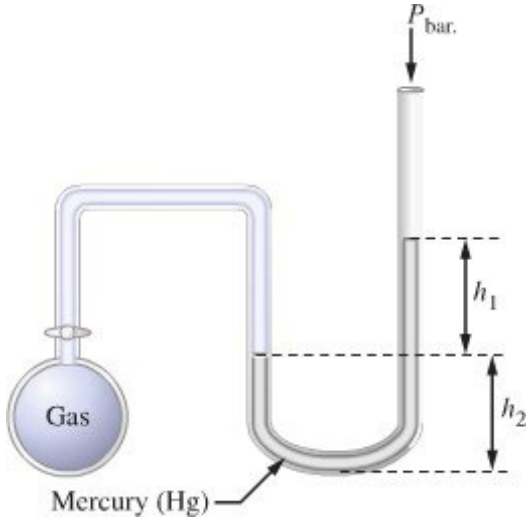


(c) Gas pressure less than barometric pressure

Refers to: pressure exerted by the height of mercury column:
1 atm = 1.01325 bar = 101.325 kPa = 760 mmHg

Question 1

- What is the pressure (in mmHg) of the gas inside the apparatus below? Given that P_{bar} = 755 mmHg, h_1 = 80 mm, h_2 = 65 mm.



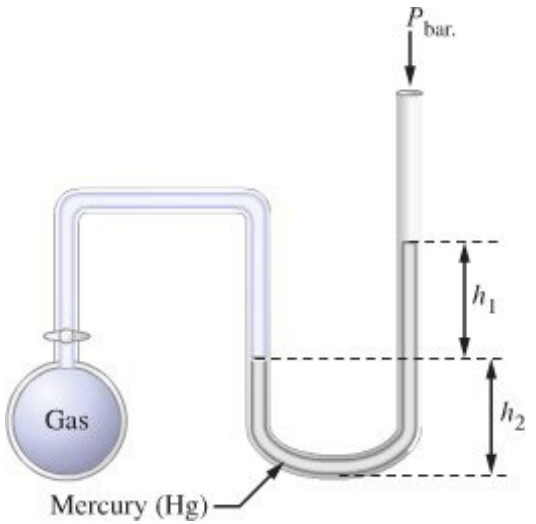
Question 1 answer

- What is the pressure (in mmHg) of the gas inside the apparatus below? Given that P_{bar} = 755 mmHg, h_1 = 80 mm, h_2 = 65 mm.

The gas pressure is greater than the barometric pressure, hence,

$P_{gas} = P_{bar} + \Delta P$

P_{gas} = 755 + 80 = 835 mmHg



The Simple Gas Laws

Boyle's Law

$PV = \text{constant}$

$P \propto \frac{1}{V}$

At constant n & T

Charles's Law

$\frac{V}{T} = \text{constant}$

$V \propto T$

At constant n & P

Avogadro's Law

$\frac{V}{n} = \text{constant}$

$V \propto n$

At constant P & T

* Express T in K in calculations

- Standard condition

$P = 1\ \text{bar} = 10^5\ \text{Pa}$

$T = 0\ ^\circ\text{C} = 273.15\ \text{K}$

- Molar volume

1 mol gas = 22.711 L (at STP)

1 mol gas = 22.414 L (at 0 °C, 1 atm)

Ideal Gas Equation, General Gas Equation and Applications

- Ideal Gas Equation: $PV = nRT$

- General Gas Equation: $\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$

- Determining molar mass: $M = \frac{mRT}{PV}$

- Determining density: $d = \frac{m}{V} = \frac{MP}{RT}$

Always Use temperature in K

Five Common Values of *R* (in various units)

$R = \frac{PV}{nT}$

$0.082057\ \text{atm L mol}^{-1}\ \text{K}^{-1}$
$0.083145\ \text{bar L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{kPa L K}^{-1}\ \text{mol}^{-1}$
$8.3145\ \text{Pa m}^3\ \text{mol}^{-1}\ \text{K}^{-1}$
$8.3145\ \text{J mol}^{-1}\ \text{K}^{-1}$

Mixture of Gases

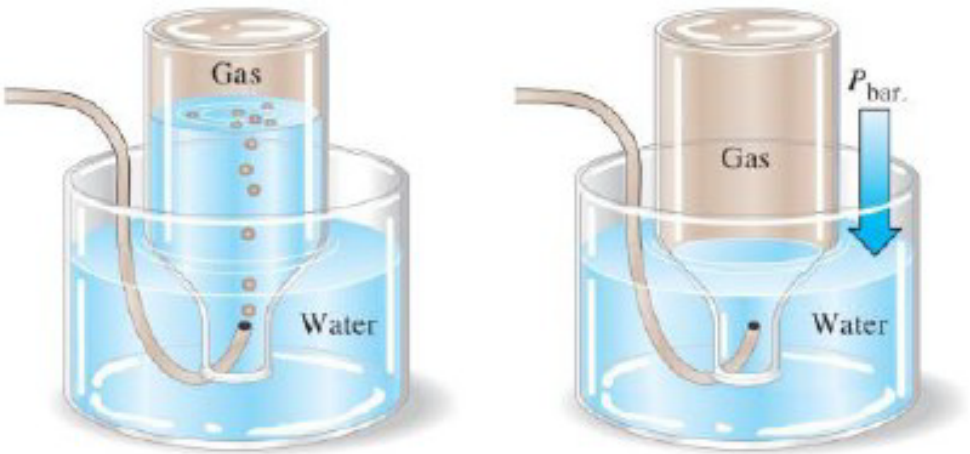
Dalton's law of partial pressure

The pressure of each gas is proportional to the number of moles of gas. The total pressure is the sum of the partial pressure of the individual gases.

$$P_{tot} = P_A + P_B + \ldots$$

$$\frac{n_A}{n_{total}} = \frac{P_A}{P_{total}} = \frac{V_A}{V_{total}} = x_A$$

Collecting a gas over water



When the position of the bottle must be adjusted so that the water levels inside and outside the bottle are the same, total gas pressure in the bottle equal to barometric pressure,

$$P_{total} = P_{bar} = P_{gas} + P_{water}$$

Atmospheric pressure (aka: Barometric p.)

Partial pressure of gas, Vapour pressure of water

Question 2

$$R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$$
$$= 0.08206\ \text{dm}^3\ \text{atm K}^{-1}\ \text{mol}^{-1}$$
$$= 0.08314\ \text{dm}^3\ \text{bar K}^{-1}\ \text{mol}^{-1}$$
$$= 8.314\ \text{m}^3\ \text{Pa K}^{-1}\ \text{mol}^{-1}$$

$M_{Ne} = 20.18$ g/mol

- At what temperature will a 20.8 g Ne sample exert a pressure of 0.818 bar when confined in a 3.20 L container?

Question 2 answer

- At what temperature will a 20.8 g Ne sample exert a pressure of 0.818 bar when confined in a 3.20 L container?

$$PV = nRT$$

$$T = \frac{PV}{nR} = \frac{0.818\ \text{bar} \times 3.20\ \text{L}}{\frac{20.8\ \text{g}}{20.18\ \text{g mol}^{-1}} \times 0.083145\ \text{bar L mol}^{-1}\ \text{K}^{-1}}$$

$$= 30.5\ \text{K}$$

Question 3

- With the help of the ideal gas equation, calculate the following at 25 $^{\circ}$C and 1 bar.

  (a) The volume of one mole of an ideal gas.

  (b) The dimension of a cube that holds one mole of ideal gas.

  (c) The density of air (in kg/m^3), given that the average molar mass of air is 28.96 g/mol.

Question 3 answer

(a) The volume of one mole of an ideal gas.

$PV = nRT$

$1\ \text{bar} \times V = 1\ \text{mol} \times 0.083145\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times (25 + 273.15)\ \text{K}$

$V = 24.8\ \text{L}$

Question 3 answer

(b) The dimension of a cube that holds one mole of ideal gas.

Let r be the length of each side of the cube.

$r^3 = 24.8\ L = 24.8 \times 1000\ cm^3$

$r = 29.2\ cm$

Question 3 answer

(c) The density of air (in kg/m^3), given that the average molar mass of air is 28.96 g/mol.

$$d = \frac{m}{V}$$

$$= \frac{\frac{28.96\ \text{g}}{1000}}{24.8\ \text{L} \times 10^{-3}}$$

$$= 1.17\ \text{kg/m}^3$$

Question 4

- A particular gaseous hydrocarbon that is 82.7% C and 17.3% H by mass has a density of 2.33 g/L at 23 $^{\circ}$C and 746 mmHg. What is the molecular formula of this hydrocarbon?

Question 4 answer

- A particular gaseous hydrocarbon that is 82.7% C and 17.3% H by mass has a density of 2.33 g/L at 23 °C and 746 mmHg. What is the molecular formula of this hydrocarbon?

Assume total mass of hydrocarbon is 100g.

	C	H
Mass	82.7g	17.3g
Mole	$\frac{82.7}{12.0} = 6.89$	$\frac{17.3}{1.0} = 17.3$
Ratio	1	$\frac{17.3}{6.89} = 2.5$
Whole number ratio	2	5

The empirical formula is C_2H_5.

For 1 L of sample, mass = 2.33g

$$PV=nRT$$

$$\frac{746}{760} \times 1 = n \times 0.082057 \times (23 + 273)$$

$$n = 0.0404 \text{ mol}$$

$$\text{molar mass} = \frac{2.33}{0.0404} = 57.7$$

Let $(C_2H_5)_n$ be the molecular formula

$$(12.0 \times 2 + 1.0 \times 5) \times n = 57.7$$

$$n = 2$$

Ans: C_4H_{10}

Question 5

- 2.0 L of O_2 (g) and 8.0 L of N_2 (g), each at 0.00 °C and 1.00 atm, are mixed together. The nonreactive gaseous mixture is compressed to occupy 2.0 L at 298 K. What is the pressure exerted by this mixture?

Question 5 answer

- 2.0 L of O_2 (g) and 8.0 L of N_2 (g), each at 0.00 °C and 1.00 atm, are mixed together. The nonreactive gaseous mixture is compressed to occupy 2.0 L at 298 K. What is the pressure exerted by this mixture?

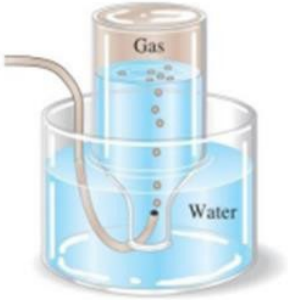
$$V_{total} = V_{O2} + V_{N2} = 10\ \text{L}$$

$$\frac{P_i V_i}{n_i T_i} = \frac{P_f V_f}{n_f T_f}$$

$$P_f = \frac{P_i V_i T_f}{V_f T_i} = \frac{1\ \text{atm} \times 10\ \text{L} \times 298\ \text{K}}{2\ \text{L} \times 273.15\ \text{K}} = 5.45\text{atm}$$

Question 6

- A sample of $MgCO_3$ was reacted with excess amount of HCl (aq) and the gas evolved was collected over water at 25 °C using the set-up below. The volume of gas collected was 75 cm^3 and the total pressure was 0.38 bar.



(a) Write down the balanced equation of the reaction.

(b) Calculate the amount of $MgCO_3$ consumed in the reaction. Given the vapor pressure of water at 25 °C is 23.8 mmHg.

Vapour Pressure of Water at Various Temperatures

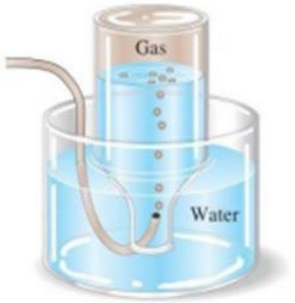
	Vapor Pressure	
t, °C	kPa	mmHg
15.0	1.706	12.79
17.0	1.938	14.54
19.0	2.198	16.49
21.0	2.448	18.66
23.0	2.811	21.08
25.0	3.170	23.78

Question 6 answer

(a) Write down the balanced equation of the reaction.

$MgCO_3(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2O(l) + CO_2(g)$

Question 6 answer



(b) Calculate the amount of $MgCO_3$ consumed in the reaction. Given the vapor pressure of water at 25 °C is 23.8 mmHg.

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$P_{total} = 0.38 \div 1.01325 \times 760 = 285$ mmHg

$P_{CO2} = 285 - 23.8 = 261$ mmHg $= 0.344$ atm

$PV = nRT$

$0.344 \times 0.075 = n \times 0.082057 \times (25 + 273)$

$n = 1.05 \times 10^{-3}$ mol

Number of moles of $MgCO_3$ = number of moles of $CO_2 = 1.05 \times 10^{-3}$ mol

Mass of $MgCO_3$ consumed $= 1.05 \times 10^{-3} \times 84.3139 = 0.0889$ g

Kinetic Molecular Theory of Gases

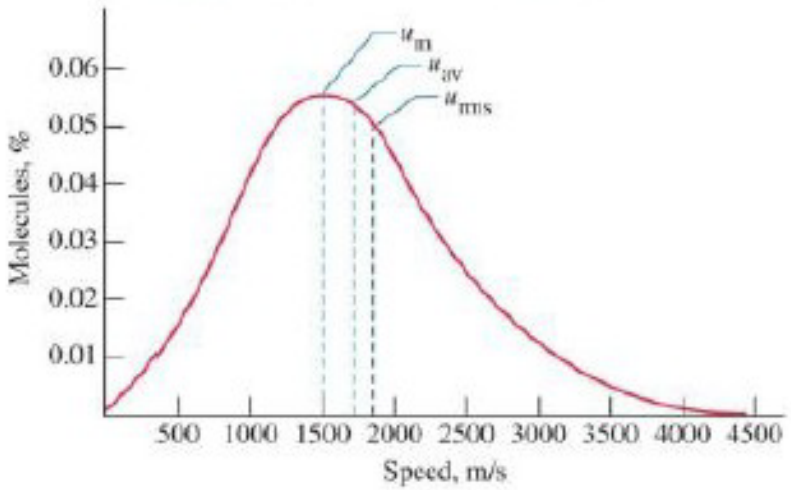
Assumptions:

- A gas is composed of a very large number of particles in constant, random, straight-line motion.
- Particles are point masses and separated by great distances.
- Collisions are rapid and elastic
- No force between particles
- Individual particles may gain or lose energy as a result of collision. But total energy remains constant.

Maxwell-Boltzmann equation:

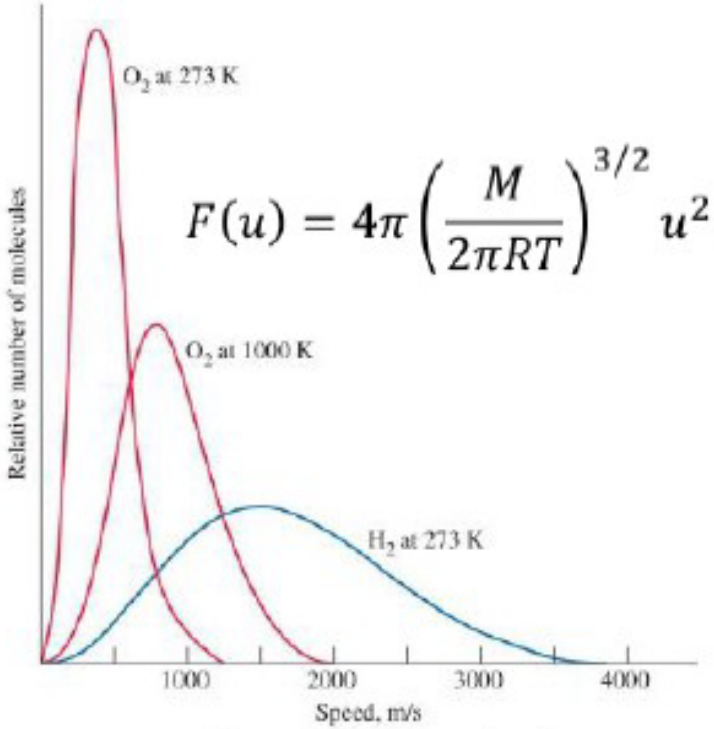
$$F(u) = 4\pi \left(\frac{M}{2\pi RT}\right)^{3/2} u^2 e^{-(\frac{Mu^2}{2RT})}$$

Distribution of molecular speeds – hydrogen gas at 0°C



$$u_m = \sqrt{\frac{2RT}{M}} \qquad u_{av} = \sqrt{\frac{8RT}{\pi M}} \qquad u_{rms} = \sqrt{\frac{3RT}{M}}$$

The distribution depends on temperature and molar mass.



$$F(u) = 4\pi\left(\frac{M}{2\pi RT}\right)^{3/2} u^2 e^{-\left(\frac{Mu^2}{2RT}\right)}$$

Area under the curve is the same when N is the same.

When T↑ or M↓, average speed ↑

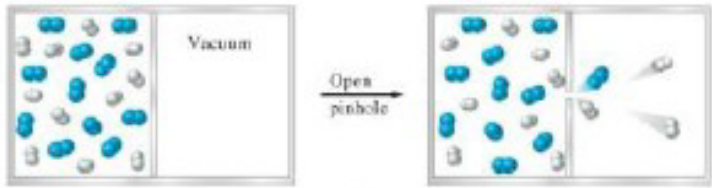
Distribution shifts towards higher speeds and become boarder.

Gas Properties Relating to the Kinetic-Molecular Theory

Diffusion: migration of molecules as a result of random motion



Effusion: escape of gas molecules from their container to vacuum



Graham's Law

Does not apply to diffusion.

$$\frac{\text{rate of effusion of A}}{\text{rate of effusion of B}} = \frac{(u_{rms})_A}{(u_{rms})_B} = \sqrt{\frac{3RT/M_A}{3RT/M_B}} = \sqrt{\frac{M_B}{M_A}}$$

Non Ideal Gas

$\frac{PV}{nRT}$ is compressibility factor.

$\frac{PV}{nRT} = 1$, ideal gas.

Prediction from gas law:
At very high pressures, $V \rightarrow 0$
But actually this cannot be.

∴ At high P, $PV_{(actual)} > PV_{(predicted)}$,
⇒ compressibility ratio > 1

Kinetic Theory of Gases assumes no intermolecular forces.
But actually there are.

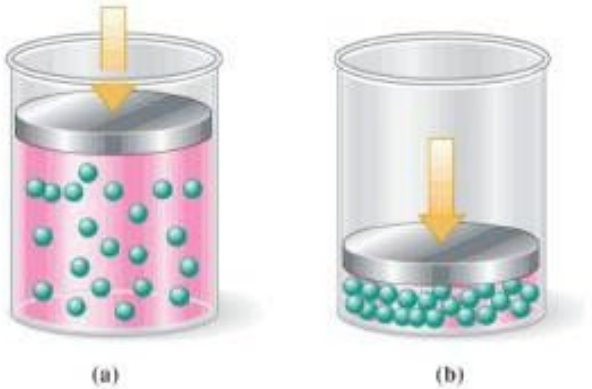
∴ At low T, $PV_{(actual)} < PV_{(predicted)}$,
⇒ compressibility ratio < 1

Van der Waal's Equation:

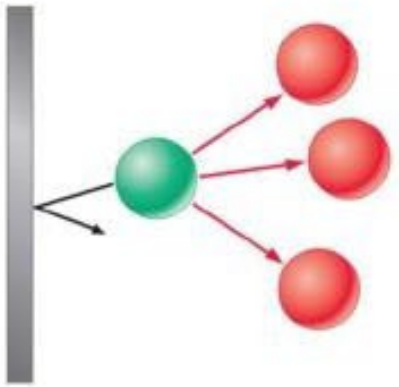
a is a measure of how strongly the molecules attract each other

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

b is called the excluded volume per mole



The effect of finite molecular size



Intermolecular forces of attraction

Attractive forces of the red molecules for the green molecule cause the green to exert less force when it collides with the wall than if these attractions did not exist.

Question 7

- Use the van der Waals equation to calculate the pressure exerted by 1.00 mol CO_2 (g) confined to a volume of 2.00 L at 273 K. Given a = 3.66 bar L^2 mol^{-2} and b = 0.0429 L mol^{-1}.

Question 7 answer

- Use the van der Waals equation to calculate the pressure exerted by 1.00 mol CO_2 (g) confined to a volume of 2.00 L at 273 K. Given a = 3.66 bar L^2 mol^{-2} and b = 0.0429 L mol^{-1}.

van der Waal equation:

$$P = \frac{nRT}{V - nb} - \frac{n^2 a}{V^2}$$

Simply substitute following values into the equation

n = 1.00 mol; V = 2.00 L; T = 273 K; R = 0.08314 bar L mol^{-1} K^{-1}

$n^2a = (1.00)^2$ mol^2 x 3.66 bar L^2 mol^{-2} = 3.66 bar L^2

nb = 1.00 mol x 0.0429 L mol^{-1} = 0.0429 L

$$P = \frac{1.00\ \text{mol} \times 0.08314\ \text{bar L mol}^{-1}\ \text{K}^{-1} \times 273\ \text{K}}{(2.00 - 0.0429)\text{L}} - \frac{3.66\ \text{bar L}^2}{2.00^2\ \text{L}^2}$$

P = 10.7 bar